

Heuristic Evaluation

Part I: Your Name

Zuhal Didem Aytac s343873

Part II: Project Description

Human.exe is a gamified diagnostic-training tool for radiology students. Users complete diagnostic challenges by examining medical images, selecting answers, or customizing their own challenge parameters.

Part III: Evaluation Execution

The evaluation was conducted in person using the group's paper prototype, with one member acting as facilitator and another as computer moving the paper screens (Giovanni Martinese and Elena Lubrano). The evaluation was performed on November 19, 2025, in Polito.

As each task was performed, I noted usability issues and tagged them with Nielsen's heuristics and severity ratings. The prototype included sufficient screens to complete all three tasks. All observations below were made directly using the paper interface.

- Simple Task: Complete a diagnostic challenge
- Moderate Task: Choose a diagnostic challenge on a specific body region
- Complex Task: Create a personalized challenge with multiple parameters

Part IV: List of Violations

1. H1 Visibility of System Status

Where: Diagnostic challenge screen

What: There is no visible timer except a vague "submit" bar acting as progress.

Why: Users cannot clearly understand how much time they have left.

Severity: 3

2. 2H1 Visibility of System Status

Where: First diagnostic challenge screen

What: The user lands on the screen after selecting a challenge, but the diagnostic question ("What pathology do you suspect?") is not visible until the next screen.

Why: Users are unsure what they are supposed to do when they arrive on the page, creating confusion about the next step.

Severity: 3

3. H1 Visibility of System Status

Where: Diagnostic challenge flow

What: The interface does not show progress (e.g., question number, total questions).

Why: Users cannot understand where they are in the challenge, reducing clarity and motivation.

Severity: 2

4. H2 Match Between System and Real World

Where: Diagnostic image display

What: The image lacks clinical context (age/gender/symptoms).

Why: Diagnosing without context is unrealistic and inconsistent with real radiology practice.

Severity: 3

5. H2 Match Between System and Real World

Where: Body region selection screen

What: The label under "Chest" lists "Pneumonia, Lung Lesions," but it is unclear whether these are examples, difficulty hints, or required knowledge.

Why: Ambiguous labeling forces users to guess the meaning and creates confusion about what they are actually selecting.

Severity: 2

6. H2 Match Between System and Real World

Where: Body region selection & personalized challenge setup

What: The lists of possible pathologies or regions are broad, unstructured, and not grouped in any meaningful way. For a student still learning, the interface provides no guidance or categorization, making selections confusing.

Why: Users must rely on prior medical knowledge or guesswork to choose; the system does not reflect how medical content is taught or organized in real learning contexts.

Severity: 3

7. H3 User Control and Freedom

Where: Diagnostic quiz

What: There is no way to go back to a previous question.

Why: Users are unable to go back to a previous question if they want to check again.

Severity: 2

8. H3 User Control and Freedom

Where: Home screen and post-challenge screen

What: Users cannot revisit previously completed challenges; the prototype provides no path back to past items.

Why: Radiology learners need to re-check past decisions, but the system gives no way to review or undo progress.

Severity: 2

9. H4 Consistency and Standards

Where: Back / Exit icons

What: Some screens have an "X" (exit), others have a back arrow, some have both, with inconsistent placement.

Why: Inconsistent icon use confuses the user about expected navigation.

Severity: 2

10. H5 Error Prevention

Where: Specific body region selection

What: The body region labels overlap the body silhouette, making it unclear what is selectable.

Why: Users may mis-tap or choose the wrong region because of ambiguous boundaries.

Severity: 2

11. H6 Recognition Rather Than Recall

Where: Personalized challenge setup

What: Difficulty, modality, and time-limit selections require remembering previous options; no explanations are provided.

Why: Users must recall what each modality implies without seeing examples.

Severity: 2

12. H7 Flexibility and Efficiency of Use

Where: Custom challenge screen

What: No presets, no "save profile," no shortcuts for frequently used parameters.

Why: Returning learners cannot quickly recreate their preferred learning mode.

Severity: 2

13. H8 Aesthetic and Minimalist Design

Where: Body region selection screen

What: Disease examples appear directly under each body region, mixing two types of information (anatomy and pathology) in a single block.

Why: Combining categories reduces visual clarity and makes it harder for students to scan the page.

Minimalist design would separate or simplify this information.

Severity: 2

14. H9 Help Users Recover from Errors

Where: Exit Challenge popup

What: "Confirm" and "Cancel" do not clearly indicate what the user will lose or keep.

Why: Users cannot recover confidently from accidental exits.

Severity: 3

15. H10 Help and Documentation

Where: Entire prototype

What: No instructional hints for first-time users (how to zoom, what modalities mean, what difficulty changes).

Why: Some guidance is necessary.

Severity: 2

16. HN Non-Heuristic Issue

Where: Correct/Wrong screens

What: Feedback text is minimal (Congrats!, You missed...) and does not support learning.

Why: Conceptual issue - the educational purpose is underdeveloped.

Severity: 2

Part V: Summary and Recommendations

Report in the table below the total number of identified violations.

Heuristic	# violations
H1: Visibility of system status	3
H2: Match between system and the real world	3
H3: User control and freedom	2
H4: Consistency and standards	1
H5: Error prevention	1
H6: Recognition rather than recall	1
H7: Flexibility and efficiency of use	1
H8: Aesthetic and minimalist design	1
H9: Help users recognize, diagnose, and recover from errors	1
H10: Help and documentation	1
HN: Non-heuristic issue	1

Overall, the prototype shows the general flow of a radiology training tool, and the main screens (home, challenge selection, diagnostic flow) are understandable. Still, many issues relate to clarity and guidance. Users often need to guess what the interface expects from them or what the medical labels mean.

A recurring problem is the lack of progress visibility: users cannot see how many questions they have, how much time is left, or where they are in the challenge. This creates uncertainty and makes the activity harder than it should be. Similarly, the medical images lack basic clinical context, which makes the diagnostic task feel unrealistic for radiology students.

Navigation also presents some friction, there is no back option, exit consequences are unclear, and past challenges cannot be revisited. These issues limit user control and slow down learning.

Recommendations

- Make progress and status more visible (time, steps, question count).
- Use clearer labels that reflect how radiology is taught in real practice.
- Give users clearer ways to recover from mistakes or explore freely (back, review, safe exit).
- Simplify the body-region screen by separating anatomical areas from example pathologies.

In general, the prototype has a solid idea behind it, but it would benefit from clearer guidance, more realistic information, and more transparent navigation to support students better.